Bachelor Thesis EDAR

EDAR V370A: An Integrative Analysis of Evolutionary Conservation, Positive Selection, and Structural Tolerance

Planned Structure Overview:

Overview	Focus
Abstract (English)	
Zusammenfassung (German)	
Table of Contents	
1. Introduction	1 EDAR background (function, signaling, and disease) 2 V370A (discovery, phenotypic effects, and geographical) **3 Research Gap: about an integrative analysis** 4 Research question, hypothesis, and overview structure
2. Comparative Genomics: Evolutionary Conservation	1 Data and methods 2 Cross-species alignment and domain architecture 2.1 Conservation profiling of the Death Domain and position 370 3 Purifying selection dN/dS 4 Synteny analysis confirming orthologs
3. Population Genetics: Detecting the V370A Sweep	1 Data, variant set, and methods 2 Allele frequency distribution of V370A 3 Population differentiation (FST) 4 Haplotype-based selection test (iHS) 5 Correlation analysis: phyloP × FST × AlphaMissense Ergebnis mit eigenem Variant-Set durchführen für individuellen Kontext und für Korrelationsanalyse. Nicht Literaturbasiert, aber als bestätigung, dass es richtig ist o.ä.
4. Structural Context: Tolerance and functional Consequence	1 Data and methods 2 Position 370 in Death Domain: localization and physicochemical properties 3 Folding stability of V370A (ΔΔG) 4 EDARADD interaction interface
5. Discussion	1 Integrative interpretation: conservation, selection, and structural tolerance 2 Phenotypic consequences and proposed selective pressure 3 Comparison with literature 4 Limitations and outlook
6. Conclusion	
7. Reference	
8. Appendix	

Working research question: Under what evolutionary constraints and structural conditions can a substitution at a conserved Death Domain position (EDAR V370A) be positively selected in East Asian populations without disrupting the core receptor function?